SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 3

Code of Conduct

I A.Laksmi(192373066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:*Lakshmi*

Assessment Tasks - Case Study

1. **Case Study 1: Smart Retail Inventory Management System**

   A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.
2. **Case Study 2: Cloud-Based Document Collaboration System**

   An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.
3. **Case Study 3: Online Food Ordering Platform**

   A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

1.Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

Introduction: A Smart Retail Inventory Management System is a cloud-based solution that helps retail stores monitor and manage inventory across multiple branches. Instead of storing data on local computers, all information is maintained on cloud servers, allowing every store to access the latest inventory details in real time. This reduces manual work, minimizes stock shortages, and improves overall business efficiency.

System Architecture

- The system consists of several components working together:
- Clients: Point of Sale (POS) systems, inventory management dashboards, and mobile applications used by employees.
- Cloud Services: These process inventory updates, sales transactions, and customer orders.
- Cloud Storage: Stores product details, supplier information, sales records, and transaction logs.
- Platform Services: Provide analytics, demand forecasting, and reporting tools.
- Network Infrastructure: VPNs, firewalls, and API gateways ensure secure communication between stores and cloud services.

Working Process:

When a customer purchases a product, the POS system immediately sends the transaction details to the cloud through an API. The cloud updates the available stock and stores the transaction record. Managers at different locations can instantly see the updated inventory through their dashboards.

The analytics platform studies historical sales data, seasonal trends, and customer demand to predict future stock requirements. This helps retailers maintain optimal inventory levels and avoid both overstocking and stock shortages.

Cloud Technologies Used

- Infrastructure as a Service (IaaS) for virtual servers and networking.
- Platform as a Service (PaaS) for analytics and forecasting applications.
- Cloud storage for product databases and transaction records.
- REST APIs for communication between clients and cloud services.

Security Features:

- VPN connections protect communication between stores and cloud servers.
- Firewalls prevent unauthorized network access.
- API Gateways authenticate client requests.
- Role-Based Access Control (RBAC) restricts access according to employee responsibilities.
- Data encryption protects sensitive customer and business information.

Advantages:

- Real-time inventory tracking.
- Improved demand forecasting.
- Reduced inventory losses.
- Easy expansion to new store locations.
- Lower maintenance costs compared to traditional systems.

Challenges:

Internet connectivity is essential.

Initial migration from legacy systems may be complex.

Continuous monitoring is required to maintain security.

2.Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

Introduction:

Modern organizations require employees to work together regardless of their location. A Cloud-Based Document Collaboration System allows multiple users to create, edit, review, and share documents simultaneously using a web browser or mobile application. Similar to Google Docs, this system improves teamwork and productivity by eliminating the need for multiple document copies.

System Architecture:

The collaboration platform includes:

Clients: Web browsers, desktop software, and mobile applications.

Web APIs: Synchronize document changes among connected users.

Cloud Storage: Stores documents, version history, comments, and backups.

Application Servers: Handle editing requests, user authentication, and synchronization.

Global Network Infrastructure: Ensures low latency and continuous availability.

Working Process:

A user opens a document through the browser and begins editing. Every change is sent through APIs to the cloud server. The server updates the document and immediately shares those changes with every connected user.

Version control automatically records every modification. If a user accidentally deletes content, earlier versions can easily be restored. Automatic cloud backups protect documents against data loss.

Cloud Technologies Used:

- SaaS (Software as a Service) for document editing.
- Distributed cloud storage for high availability.
- Web APIs for real-time synchronization.
- Content Delivery Networks (CDNs) for faster global access.

Security Features:

- End-to-end encryption protects document data.
- Identity management verifies user identities.
- Access permissions control who can view or edit documents.
- Multi-factor authentication provides additional account security.
- Automatic backup and disaster recovery ensure data protection.

Advantages:

- Supports real-time collaboration.
- Accessible from any internet-connected device.
- Eliminates duplicate document versions.
- Automatic saving and backup.
- Easy sharing with internal and external users.

Challenges:

- Requires stable internet connectivity.
- Managing permissions becomes difficult in large organizations.
- Security policies must be updated regularly.

3.Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Introduction: Online food ordering platforms have become an essential part of modern life. Customers can browse restaurant menus, place orders, make online payments, and track deliveries through websites or mobile applications. Cloud computing provides the scalability, availability, and security required to handle thousands of users simultaneously.

System Architecture:

- The platform consists of:
- Frontend Clients: Mobile applications and web browsers used by customers.
- Backend Services: Process orders, payments, customer information, and restaurant data.
- RESTful APIs: Connect frontend applications with backend services.
- Cloud Storage: Stores menu images, invoices, customer profiles, and order history.
- Infrastructure: Virtual machines and Kubernetes clusters manage application deployment and scaling.

Working Process:

A customer logs into the application, selects food items, and places an order. The frontend sends the request to backend services through REST APIs. The backend validates payment, stores the order in cloud databases, and forwards it to the restaurant.

The restaurant accepts the order, and customers receive real-time status updates until delivery is completed. Kubernetes automatically scales application resources during busy periods, ensuring consistent performance.

Cloud Technologies Used:

- IaaS for virtual machines.
- Kubernetes for container orchestration.
- Cloud storage for images and customer data.
- RESTful APIs for communication.
- Database services for order management.

Security Features:

- HTTPS encrypts all communication.
- Authentication tokens verify user sessions.

- Role-Based Access Control (RBAC) limits access for customers, restaurant staff, and administrators.
- Secure cloud storage protects personal information.
- Regular backups ensure business continuity.

Advantages:

- High scalability during peak demand.
- Fast response time.
- Secure online transactions.
- Easy deployment of new features.
- High availability with minimal downtime.

Challenges:

- Managing high traffic during festivals and weekends.
- Protecting customer payment information.
- Monitoring application performance continuously.
- Maintaining reliable API communication.